

Customer Retention & Churn Analysis Report

Executive Summary

Customer retention is a critical factor influencing revenue growth and long-term business sustainability. This report presents an analysis of customer churn behavior using the Telco Customer Churn Dataset. The objective is to identify high-risk customer segments, understand factors contributing to churn, and recommend actionable strategies to improve customer retention.

The analysis revealed an overall churn rate of **26.54%**, indicating that approximately one in four customers discontinued their services. Key drivers of churn include month-to-month contracts, fiber optic internet services, electronic check payment methods, and senior citizen customer segments.

1. Introduction

Customer churn refers to the loss of customers over a specific period. Understanding churn patterns helps organizations reduce customer attrition, improve customer satisfaction, and increase profitability.

Project Objectives

- Analyze customer churn trends and retention behavior.
 - Identify customer segments with higher churn risk.
 - Evaluate the impact of contracts, payment methods, and internet services.
 - Provide data-driven recommendations to improve retention.
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2. Dataset Overview

The analysis was conducted using the Telco Customer Churn Dataset, which contains customer demographic information, account details, subscribed services, payment methods, and churn status.

Key Dataset Features

- Customer Demographics
- Contract Type
- Internet Service Type
- Payment Method
- Monthly Charges
- Tenure
- Churn Status

Tools Used

- Power BI
 - Microsoft Excel
 - Data Cleaning and Transformation
 - Business Intelligence Reporting
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3. Dashboard Overview

The dashboard consists of two major sections:

Executive Summary Dashboard

Provides a high-level overview of:

- Total Customers
- Active Customers
- Churned Customers
- Churn Rate
- Average Monthly Revenue

Customer Segmentation Dashboard

Provides insights into:

- Churn by Gender
 - Churn by Senior Citizen Status
 - Churn by Contract Type
 - Churn by Payment Method
 - Churn by Internet Service
 - Customer Tenure Analysis
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4. Key Findings

4.1 Overall Churn Performance

Metric	Value
Total Customers	7,043
Active Customers	5,174
Churned Customers	1,869

Metric	Value
Churn Rate	26.54%
Average Monthly Revenue	\$64.76

Observation

The churn rate of 26.54% suggests a significant opportunity to improve customer retention and business performance.

4.2 Contract Type Analysis

Month-to-month contract customers account for the highest proportion of churned customers.

Business Impact

Short-term contracts provide customers with greater flexibility to switch providers, increasing churn risk.

4.3 Internet Service Analysis

Customers using Fiber Optic services exhibit the highest churn levels compared to DSL users.

Business Impact

This may indicate service quality concerns, pricing issues, or customer expectations not being met.

4.4 Payment Method Analysis

Electronic Check users show the highest churn rate among all payment methods.

Business Impact

Customers using manual payment methods may be less engaged and more likely to discontinue services.

4.5 Customer Demographics

Senior Citizens experience a substantially higher churn rate than non-senior customers.

Gender has minimal influence on churn behavior.

Business Impact

Targeted retention programs may be required for senior customers.

5. Recommendations

Based on the findings, the following actions are recommended:

Recommendation 1: Promote Long-Term Contracts

Offer loyalty rewards, discounts, and bundled services to encourage customers to adopt annual contracts.

Recommendation 2: Improve Fiber Optic Service Experience

Conduct customer satisfaction surveys and investigate service quality issues affecting Fiber Optic customers.

Recommendation 3: Encourage Automatic Payments

Provide incentives for customers to switch from Electronic Check payments to automated payment methods.

Recommendation 4: Develop Senior Citizen Retention Programs

Create personalized support channels and exclusive retention offers for senior customers.

Recommendation 5: Implement Predictive Retention Strategies

Use churn indicators to identify high-risk customers and proactively engage them before cancellation.

6. Conclusion

This analysis identified several factors strongly associated with customer churn, including contract type, internet service, payment method, and customer demographics. By implementing targeted retention strategies, the organization can reduce churn, improve customer satisfaction, and increase long-term revenue growth.

The insights generated through this project demonstrate how data analytics and business intelligence can support strategic decision-making and customer retention initiatives.

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